

Lecture 11, Jan 14, 2026.

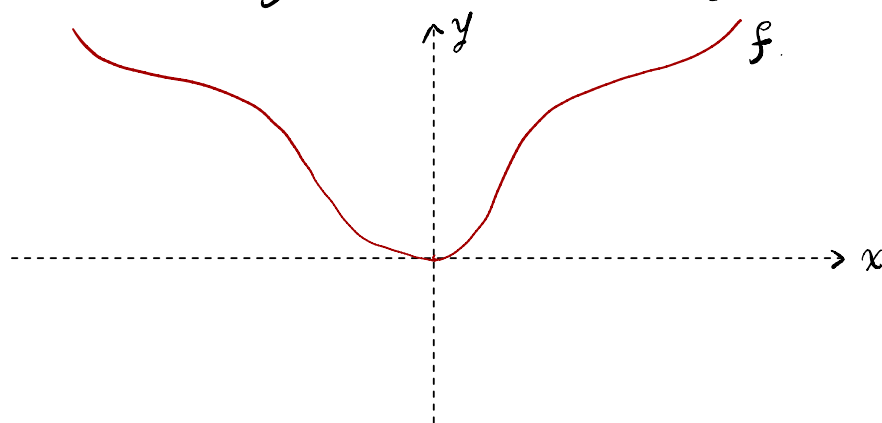
3.5 Guidelines for sketching a Curve.

A). Domain. ($\text{Dom } f$)

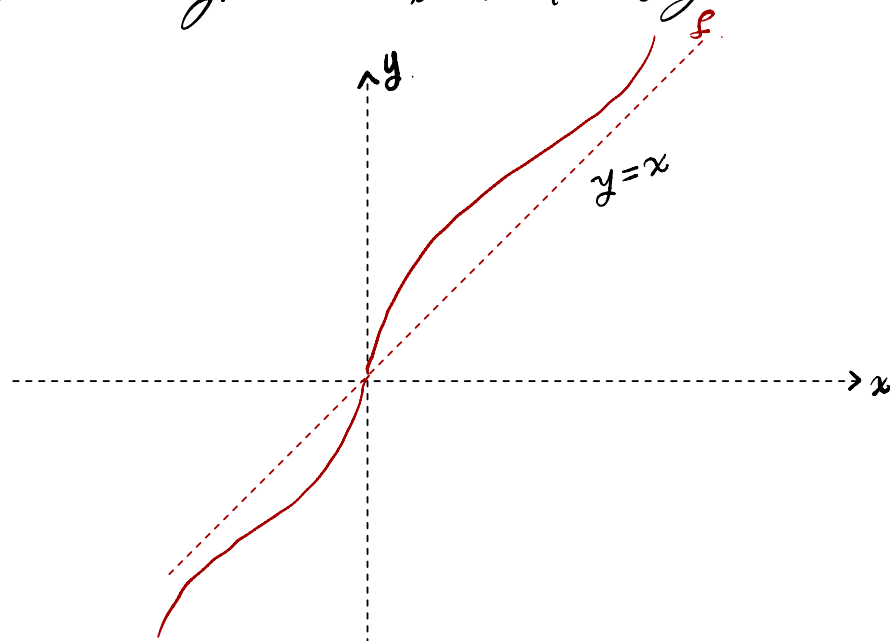
B). Intercepts.

C). Symmetry:

i). If $f(-x) = f(x)$ for all $x \in \text{Dom } f$ then f is an even function and the curve is symmetric about the y -axis.



ii). if $f(-x) = -f(x)$ for all $x \in \text{Dom } f$, then f is an odd function and the curve is symmetric about the origin.



D). Asymptotes.

i). Horizontal Asymptotes:

if $\lim_{x \rightarrow \infty} f(x) = L \Rightarrow y = L$ is HA.

if $\lim_{x \rightarrow -\infty} f(x) = K \Rightarrow y = K$ is HA.

ii). Vertical Asymptotes: The line $x=a$ is a vertical asymptote if one of the following statements is true:

$$\lim_{x \rightarrow a^+} f(x) = \infty \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty \quad \lim_{x \rightarrow a^-} f(x) = -\infty$$

iii). Slant Asymptote:

E). Intervals of Increase or Decrease: Use $f'(x) > 0$ and solve if inequality.

F). Local Maximum or Minimum values: Find the critical numbers of f where $f'(c) = 0$ or $f'(c)$ DNE. Then use the First Derivative test or Second Derivative test to compute these.

G). Concavity and Points of inflection: Compute $f''(x)$ and use.

$$f''(x) > 0 \quad \text{then } \cup$$

$$f''(x) < 0 \quad \text{then } \cap$$

and inflection points are points s.t $f''(c) = 0$ or DNE.

H). Sketch the curve:

Example. Use the guidelines to sketch the curve $y = \frac{2x^2}{x^2-1}$.

Sln. A. Domain.

$$x^2-1=0 \Rightarrow x = \pm 1 \Rightarrow \text{Dom } f = \mathbb{R} - \{\pm 1\}.$$

B. Intercepts: $x=0 \Rightarrow y = \frac{2(0)^2}{0^2-1} \Rightarrow (0,0)$ y-intercepts.

$$x\text{-intercepts: } y=0 \quad \text{then} \quad \frac{2x^2}{x^2-1} = 0 \Rightarrow x=0. \quad (0,0).$$

C). Symmetry: f is even.

$$f(-x) = \frac{2(-x)^2}{(-x^2)-1} = \frac{2x^2}{x^2-1} = f(x).$$

D. Asymptotes: Vertical: the possible vertical asymptotes are $x = \pm 1$.

$$\lim_{x \rightarrow 1^+} \frac{2x^2}{x^2-1} = \infty \quad \text{and} \quad \lim_{x \rightarrow 1^-} \frac{2x^2}{x^2-1} = -\infty.$$

$$\lim_{x \rightarrow -1^+} \frac{2x^2}{x^2-1} = -\infty \quad \text{and} \quad \lim_{x \rightarrow -1^-} \frac{2x^2}{x^2-1} = +\infty.$$

So, $x=1$, $x=-1$ are vertical asymptotes.

Horizontal. $\lim_{x \rightarrow \infty} \frac{2x^2}{x^2-1} = 2$; $\lim_{x \rightarrow -\infty} \frac{2x^2}{x^2-1} = 2$.

So, $y=2$ is a horizontal asymptote.

E. Intervals of Increase/Decrease: $f(x) = \frac{2x^2}{x^2-1}$

$$f'(x) = \frac{4x(x^2-1) - 2x^2(2x)}{(x^2-1)^2} = \frac{4x^3 - 4x - 4x^3}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$$

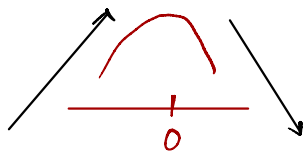
Solving $f'(x) > 0 \Leftrightarrow \frac{-4x}{(x^2-1)^2} > 0 \Leftrightarrow -4x > 0 \Leftrightarrow x < 0$

f is increasing on $(-\infty, -1) \cup (-1, 0)$

and f is decreasing on $(0, 1) \cup (1, +\infty)$.

Critical points: Possible critical points $x=0$, $x=1$, $x=-1$ but since $x = \pm 1 \notin \text{Dom } f$ then the only critical point is $x=0$.

F. Local min/max values: By the first Derivative test since f' changes from (+) to (-) at 0, $f(0)=0$ is a local maximum

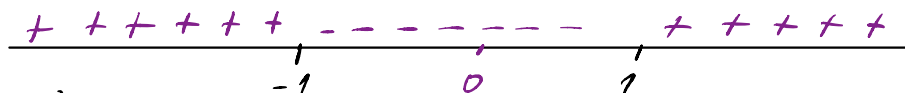


G. Concavity and points of inflection: $f'(x) = \frac{-4x}{(x^2-1)^2}$

$$\begin{aligned} f''(x) &= \frac{-4(x^2-1)^2 - (-4x)(2)(x^2-1)(2x)}{(x^2-1)^4} = \frac{\cancel{(x^2-1)}(-4(x^2-1) + 16x^2)}{(x^2-1)^4} \\ &= \frac{-4x^2 + 4 + 16x^2}{(x^2-1)^3} = \frac{12x^2 + 4}{(x^2-1)^3} \end{aligned}$$

Notice that $12x^2+4 > 0$ for $x \in \text{Dom } f$.

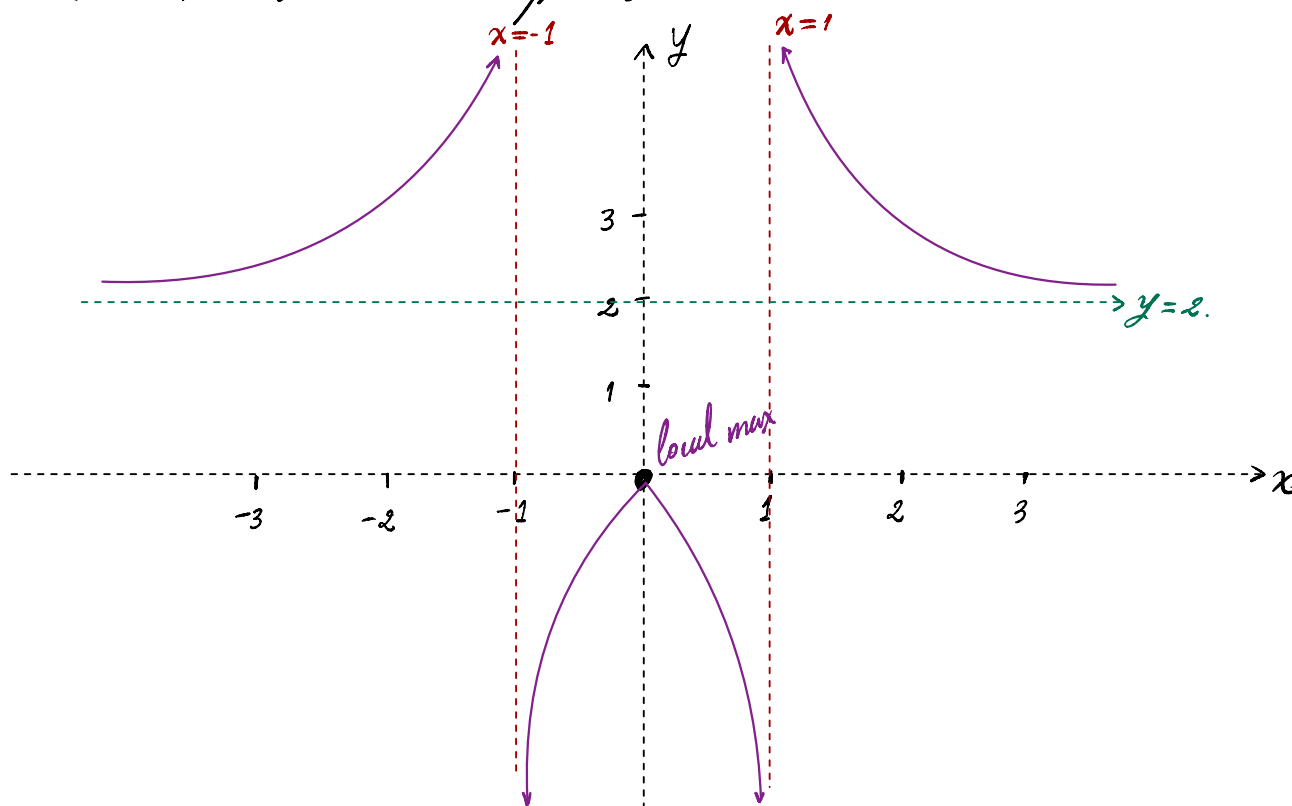
$$\text{So, } f''(x) > 0 \Leftrightarrow (x^2-1)^3 > 0 \Leftrightarrow x^2-1 > 0 \Leftrightarrow (x+1)(x-1) > 0$$



So f is $\cup$ on $(-\infty, -1) \cup (1, \infty)$

f is $\cap$ on $(-1, 1)$.

$f''(x) \neq 0$ b/c $12x^2+4 > 0$ and $f'(x)$ DNE at $x = \pm 1$. but $\pm 1 \notin \text{Dom } f$, then there is not inflection points.



Example. sketch the graph of $f(x) = \frac{x^2}{\sqrt{x+1}}$.

Sln. A. $\text{Dom } f = (-1, \infty)$.

B. the x - and y - intercepts are both $(0, 0)$.

C. Symmetry : None

$$f(-x) = \frac{(-x)^2}{\sqrt{-x+1}} = \frac{x^2}{\sqrt{1-x}} \neq f(x) \neq -f(x).$$

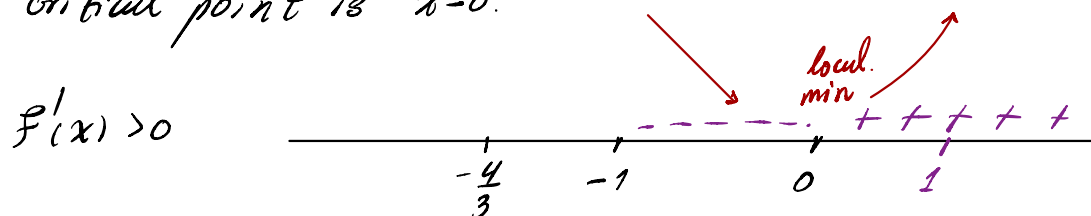
D. Asymptotes: Horizontal:

$$\lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x+1}} = \infty \text{ there is no H.A.}$$

$\lim_{x \rightarrow -1^+} \frac{x^3}{\sqrt{x+1}} = \infty$ so, $x = -1$ is a vertical asymptote.

$$E. f'(x) = \frac{2x\sqrt{x+1} - x^2\left(\frac{1}{2\sqrt{x+1}}\right)}{x+1} = \frac{4x(x+1) - x^2}{2\sqrt{x+1}(x+1)} = \frac{3x^2 + 4x}{2\sqrt{x+1}(x+1)}$$

$f'(x) = 0 \Leftrightarrow 3x^2 + 4x = 0 \Leftrightarrow x = 0, x = -\frac{4}{3} \notin \text{Dom } f$ the only critical point is $x = 0$.



So f is increasing on $(0, \infty)$ and decreasing on $(-1, 0)$.

E. So by the first derivative test $x = 0, f(0) = 0$ is local minimum

$$F. f''(x) = \frac{(6x+4)(2(x+1)^{3/2}) - x(3x+4)(3(x+1)^{1/2})}{(2(x+1)^{3/2})^2}$$

$$= \frac{2(6x+4)(x+1)^{3/2} - 3x(3x+4)(x+1)^{1/2}}{4(x+1)^3}$$

$$= \frac{(x+1)^{1/2} [2(6x+4)(x+1) - 3x(3x+4)]}{4(x+1)^3}$$

$$= \frac{2(6x^2 + 10x + 4) - 9x^2 - 12x}{4(x+1)^{5/2}}$$

$$= \frac{3x^2 + 8x + 8}{4(x+1)^{5/2}}$$

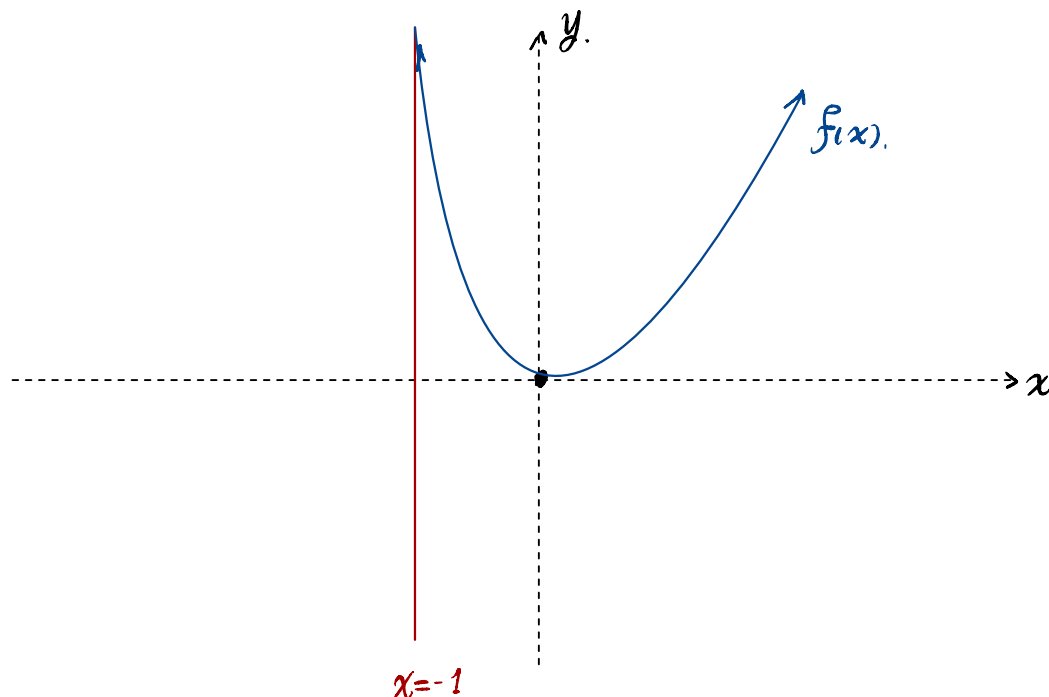
$$3x^2 + 8x + 8 = 0 \Leftrightarrow x = \frac{-8 \pm \sqrt{8^2 - 4(3)(8)}}{6} = \frac{-8 \pm \sqrt{64 - 96}}{6}$$

then $3x^2 + 8x + 8 > 0$ or $3x^2 + 8x + 8 < 0$ for all $x \in \text{Dom } f$.

$$x = 1 \in \text{Dom } f, \quad 3(1)^2 + 8(1) + 8 = 21 > 0$$

then $3x^2 + 8x + 8 > 0$ for all $x \in \text{Dom } f$.

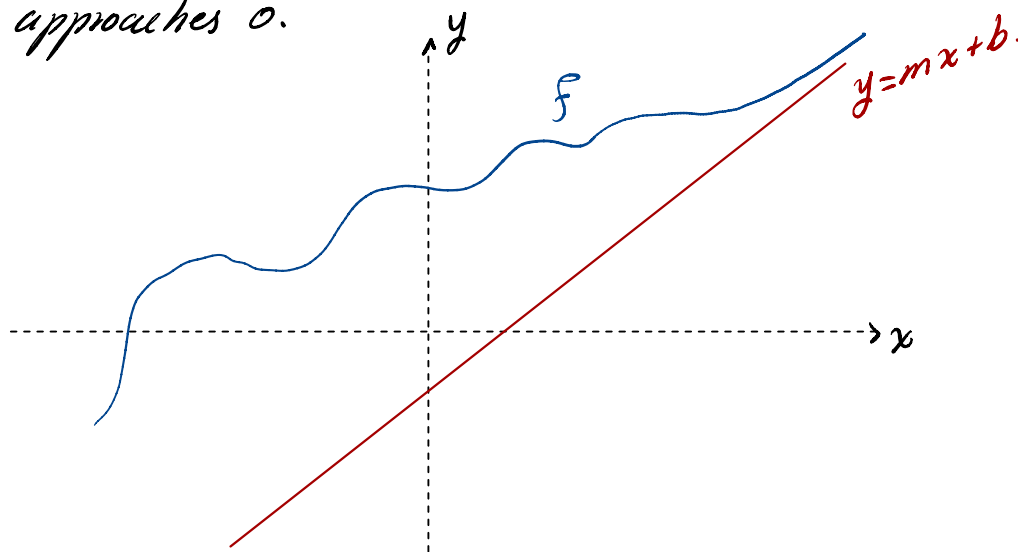
$f''(x) > 0 \Rightarrow (x+1)^{5/2} > 0 \Rightarrow \sqrt{(x+1)^5} > 0$ then,
 f is $\cup$ on $\text{Dom } f = (-1, +\infty)$.



Slant Asymptotes: Some curves have asymptotes that are oblique, that is, neither horizontal nor vertical. if

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$$

where $m \neq 0$, then $y = mx + b$ is called a slant asymptote b/c the vertical distance between the curve $y = f(x)$ and the line $y = mx + b$ approaches 0.



Example. Compute slant asymptote of $f(x) = \frac{x^3}{x^2+1}$.

Sln.

$$\begin{array}{r|l} x^3 & x^2+1 \\ \hline -x^3-x & \boxed{x} \\ \hline -x & \end{array}$$

we done if degree here is $<$ degree x^2+1 .

→ then $y=x$ is the possible slant asymptote

$$\lim_{x \rightarrow \infty} \left[\frac{x^3}{x^2+1} - x \right] \stackrel{?}{=} 0$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \left[\frac{x^3 - x^3 - x}{x^2+1} \right] &= \lim_{x \rightarrow \infty} \frac{-x}{x^2+1} \\ &= \lim_{x \rightarrow \infty} \frac{-x/x^2}{x^2/x^2 + 1/x^2} \\ &= \lim_{x \rightarrow \infty} \frac{-1/x}{1 + 1/x^2} = 0. \end{aligned}$$

then $y=x$ is a slant asymptote of $y=f(x)$.

• $\text{Dom } f = \mathbb{R}$

• x - and y - intercepts are $(0,0)$.

• $f(-x) = \frac{(-x)^3}{(-x)^2+1} = -\frac{x^3}{x^2+1} = -\frac{x^3}{x^2+1} = -f(x)$ is an odd function.

• No asymptotes.

•

$$f'(x) = \frac{3x^2(x^2+1) - x^3(2x)}{(x^2+1)^2} = \frac{x^4 + 3x^2}{(x^2+1)^2} = \frac{x^2(x^2+3)}{(x^2+1)^2}$$

f is increasing on $(-\infty, +\infty)$ $x=0$ is the only critical point.

at 0, there is not local min or max By the first Derivative test

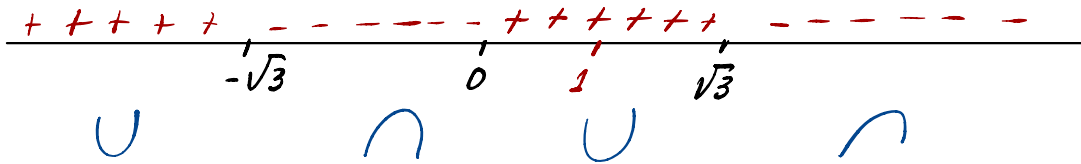
$$f''(x) = \frac{(4x^3+6x)(x^2+1)^2 - (x^4+3x^2)(2)(x^2+1)(2x)}{(x^2+1)^4}$$

$$= \frac{(x^2+1) ((4x^3+6x)(x^2+1) - 4x(x^4+3x^2))}{(x^2+1)^4}$$

$$= \frac{2x ((2x^2+3)(x^2+1) - 2x^4 - 6x^2)}{(x^2+1)^3}$$

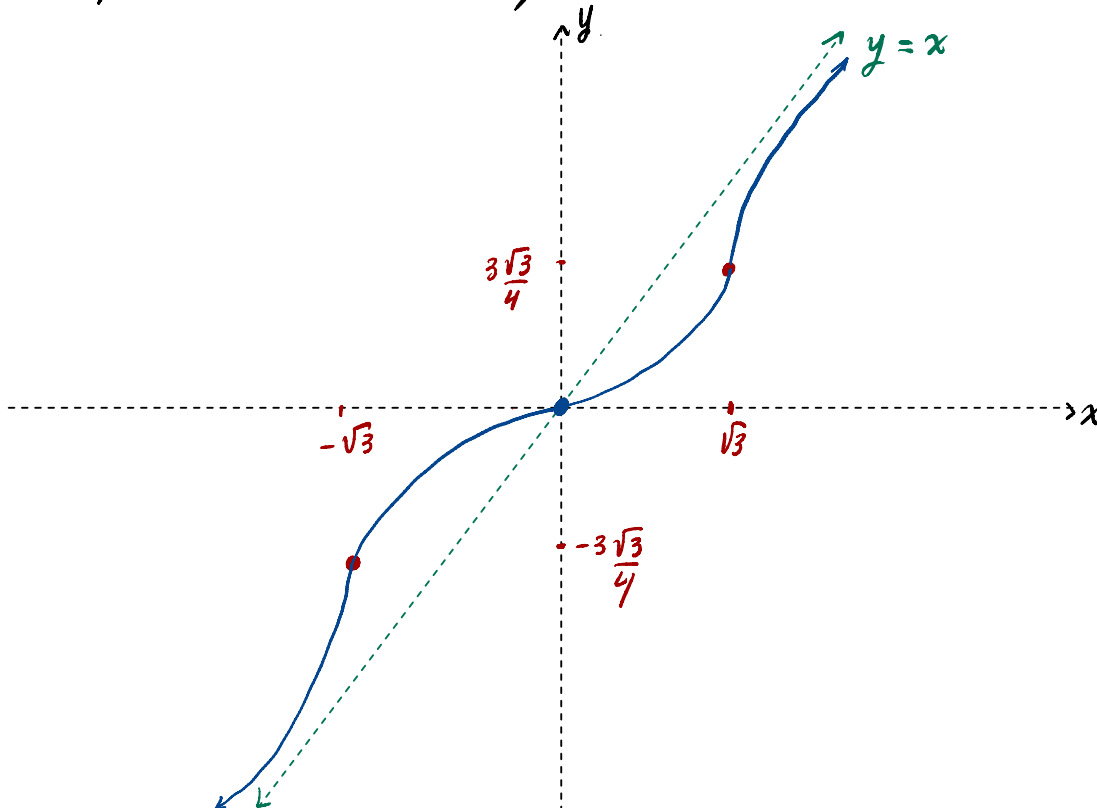
$$= \frac{2x (2x^4 + 5x^2 + 3 - 2x^4 - 6x^2)}{(x^2+1)^3}$$

$$= \frac{2x (3-x^2)}{(x^2+1)^3} \quad f''(x)=0 \Rightarrow x=0, \quad x=\pm\sqrt{3}.$$



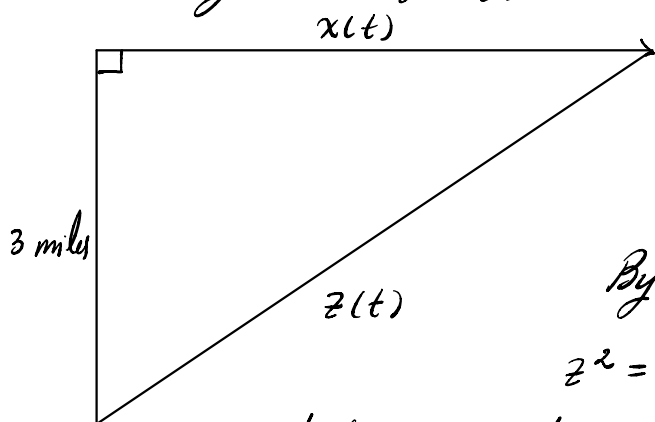
$$x=\sqrt{3}; \quad f(\sqrt{3}) = \frac{3\sqrt{3}}{4}; \quad f(-\sqrt{3}) = -f(\sqrt{3}) = -\frac{3\sqrt{3}}{4}$$

$(\sqrt{3}, \frac{3\sqrt{3}}{4})$ and $(-\sqrt{3}, -\frac{3\sqrt{3}}{4})$ are inflection points.



Applications of Implicit Differentiation.

Problem 1 A plane flying horizontally at an altitude of 3 miles and a velocity of 300 mi/hr passes directly over a radar station. Find the rate at which the straight line distance from the plane to the station is increasing when the plane's straight line distance is 5 miles away from the station.



$$\frac{dx}{dt} = 300 \text{ mi/hr.}$$

$$\frac{dz}{dt} = ? \text{ when } z = 5 \text{ miles.}$$

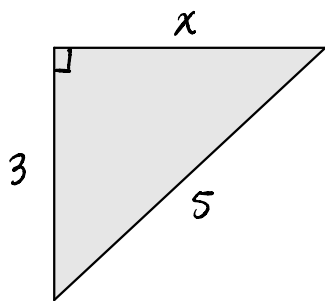
By P-thm.

$$z^2 = x^2 + 3^2$$

$$\frac{d}{dt}(z^2) = \frac{d}{dt}(x^2 + 3^2)$$

$$2z \cdot \frac{dz}{dt} = 2x \cdot \frac{dx}{dt} + 0 \quad ; \quad z=5, \quad \frac{dx}{dt} = 300$$

$$\frac{dz}{dt} = \frac{x}{z} \frac{dx}{dt}$$



$$5^2 = x^2 + 3^2 \Rightarrow x = \sqrt{25-9} = 4$$

$$\text{plug in } z=5, x=4; \frac{dz}{dt} = \frac{4}{5}(300) = 240 \text{ mi/hr.}$$